

Cybersecurity

Autopsy and iPhone Forensics

Lesson 21.3



Learning Objectives

By the end of today's lesson, you will be able to:

- 1 Identify the methods used in smartphone forensics investigations.
- 2 Navigate the database and file structure of the iPhone's flash drive.
- 3 Locate identifiable evidence on the iPhone in order to establish ownership.
- 4 Use Autopsy to view and tag evidence in an iPhone image.
- 5 Extract image content for offline viewing in other applications (logs, text, pictures, video, audio).



Welcome



Let's Recap



Recap

In the last lesson, we covered:

01

Basic principles of digital forensics methodologies.

02

Various skill sets required for a career in digital forensics.

03

Proper approach to collecting, preserving, analyzing, and reporting digital forensics evidence.

04

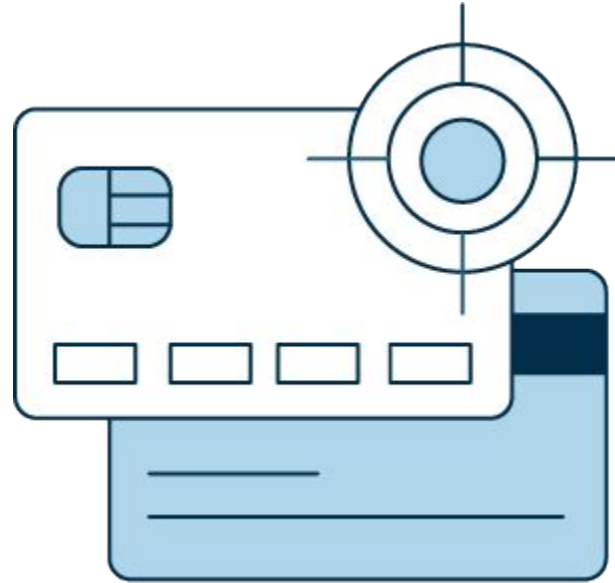
How to conduct a preliminary review for a digital forensics case.

05

How to preserve and document evidence using Autopsy.

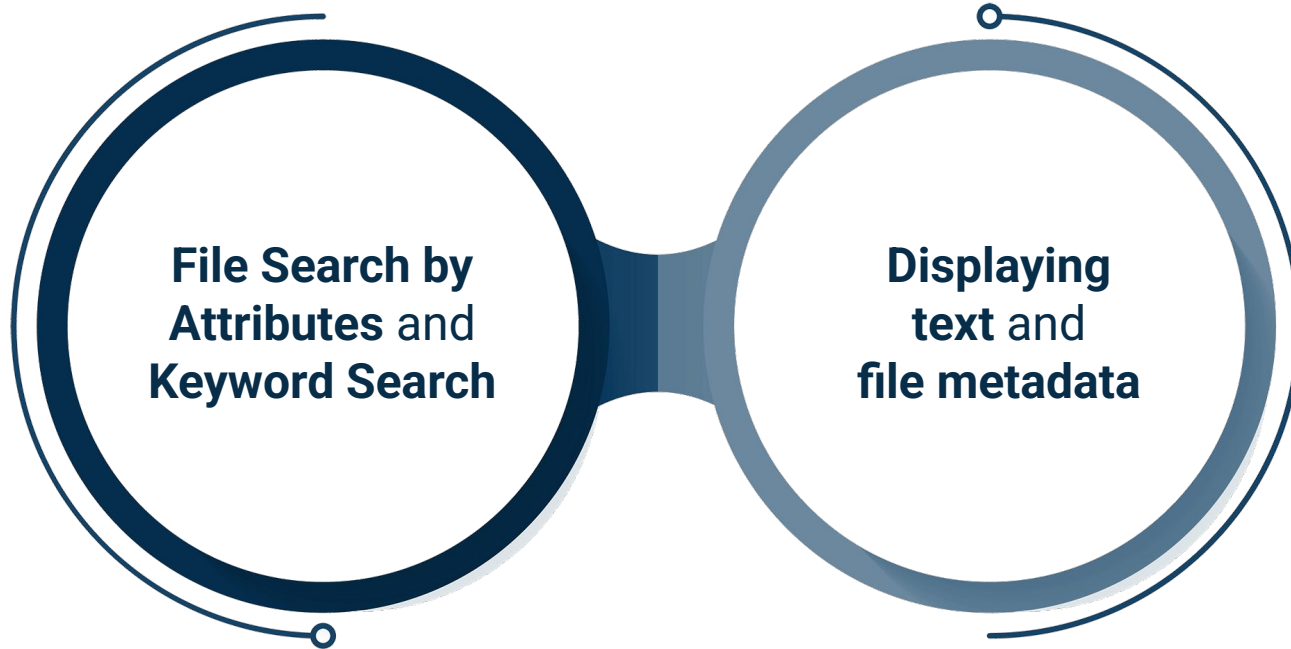
Overview

Today we will focus on smartphone forensics and continue the **Pure Gold Credit Union** case by using Autopsy to view and tag evidence on an iPhone image.



Overview

You will also learn how to perform image searches without using the directory tree.
Instead, you will use:





File Systems and Data Storage

Where's the data?

File Systems and Data Storage

Where's the data?

An iPhone does not have external storage.
All data is stored in **flash memory** on the device.

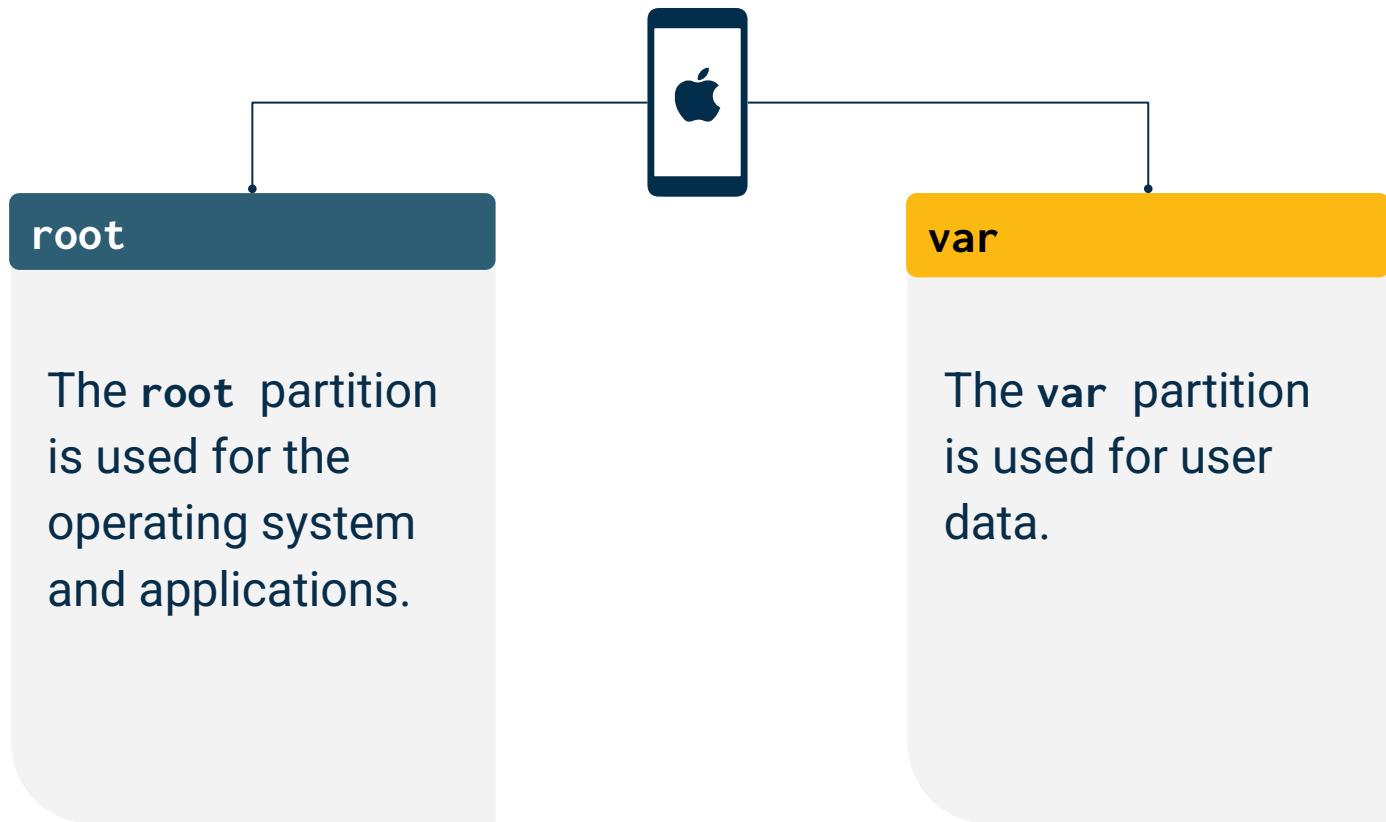
Remember

- Flash memory retains data without any external power and is used in SSDs and smartphones.
- Flash memory on an iPhone exists **until** it is overwritten.



File Systems and Data Storage

An iPhone's flash memory drive contains two disk partitions:



Data is first imaged using a **bit-level copy**.

The background of the entire image is a blue-tinted photograph of a landscape with several cell towers. Overlaid on this are several white location pin icons. A hand is holding a smartphone in the foreground, which displays two paragraphs of text.

In an iPhone, this will
recover deleted
messages as well as
GPS coordinates and
cell tower locations.

iPhones allow users to
back up their data to the
cloud, which can be
another source of data
in an investigation,
especially when the
device is not available.



What challenges
does iCloud data
present to a digital
forensics investigator?



Answer

Isolating and securing evidence is challenging when data is in multiple locations. Also, Service Level Agreements (SLAs) must be recognized when dealing with other companies, in addition to other legal issues.

Important Directories, Databases, and Files

The following directories are worth investigating for evidence:

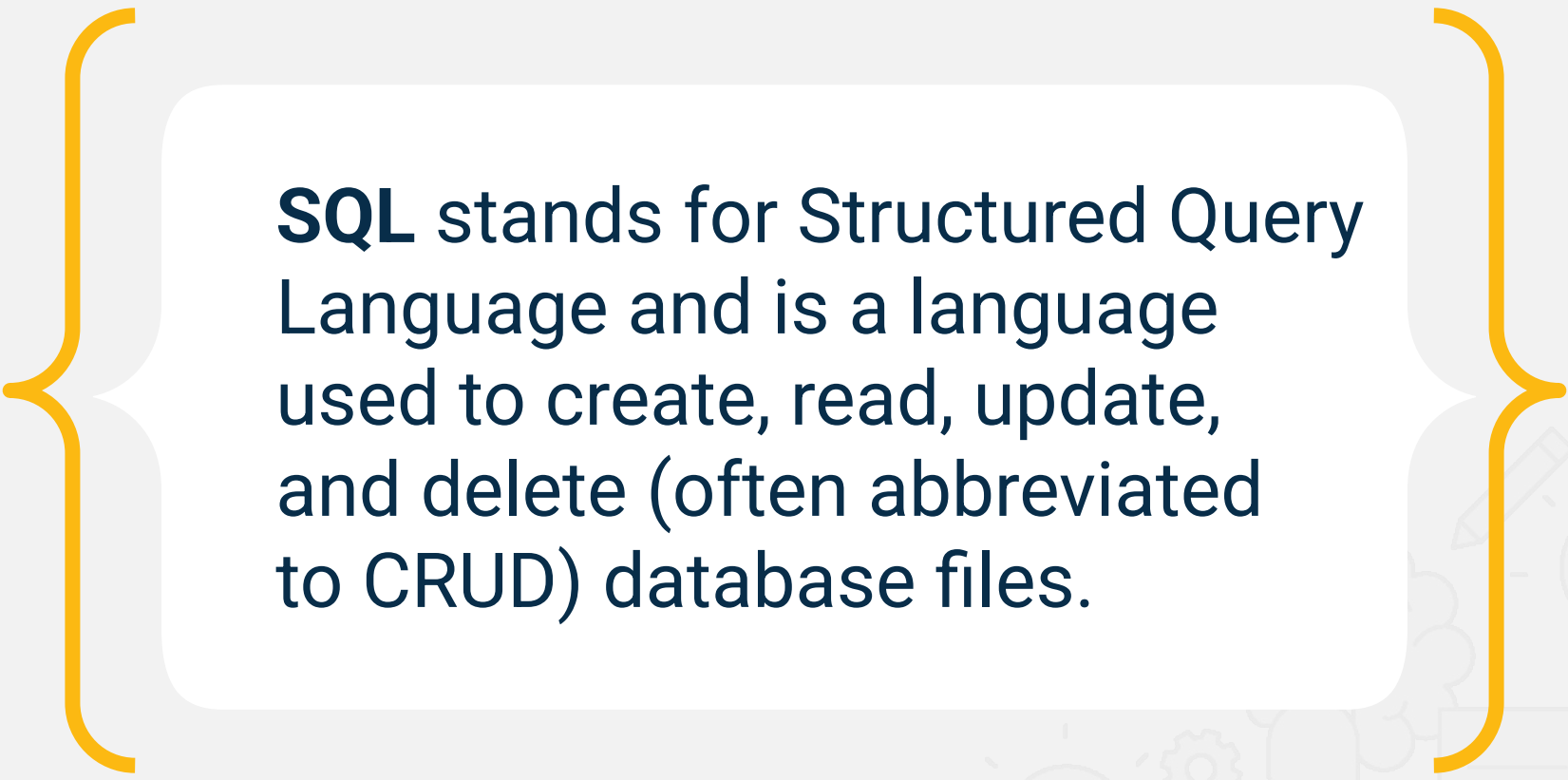


Since iPhone's iOS is based on Unix, the directory structure should be familiar.

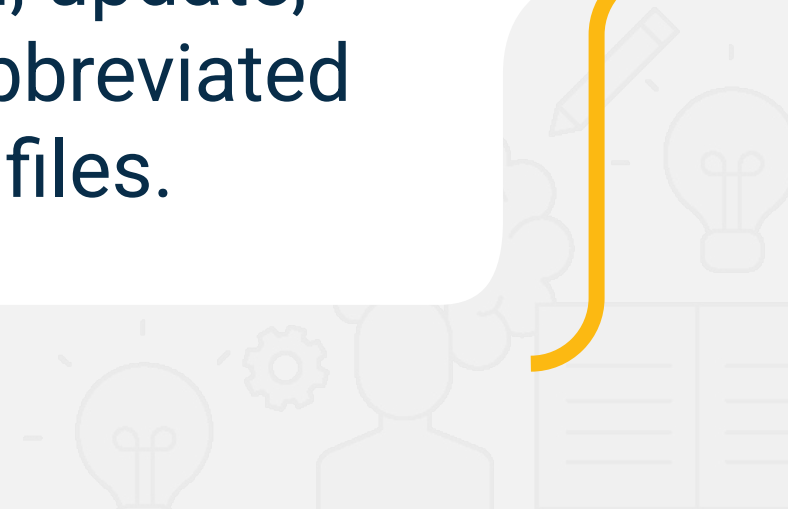
Most information is in a human-readable format within the Autopsy application.



The iPhone stores user data in **SQL** databases and other files.



SQL stands for Structured Query Language and is a language used to create, read, update, and delete (often abbreviated to CRUD) database files.



Important Directories, Databases, and Files

AddressBook.sqlitedb	Contact information and personal data like a person's name, email address, etc.
AddressBookImages.sqlitedb	Images associated with saved contacts
Calendar.sqlitedb	Calendar details and events information
CallHistory.db	Call logs including phone numbers and timestamps
sms.db	Text and multimedia messages along with timestamps
voicemail.db	Voicemail messages
Safari/Bookmarks	Saved URL addresses
locationd	Google coordinates of locations

Important Directories, Databases, and Files

iPhone also has data stored in property lists (**plists**).

activation_record.plist

Contains phone model information including ICCID, IMEI, and serial number.

data_ark.plist

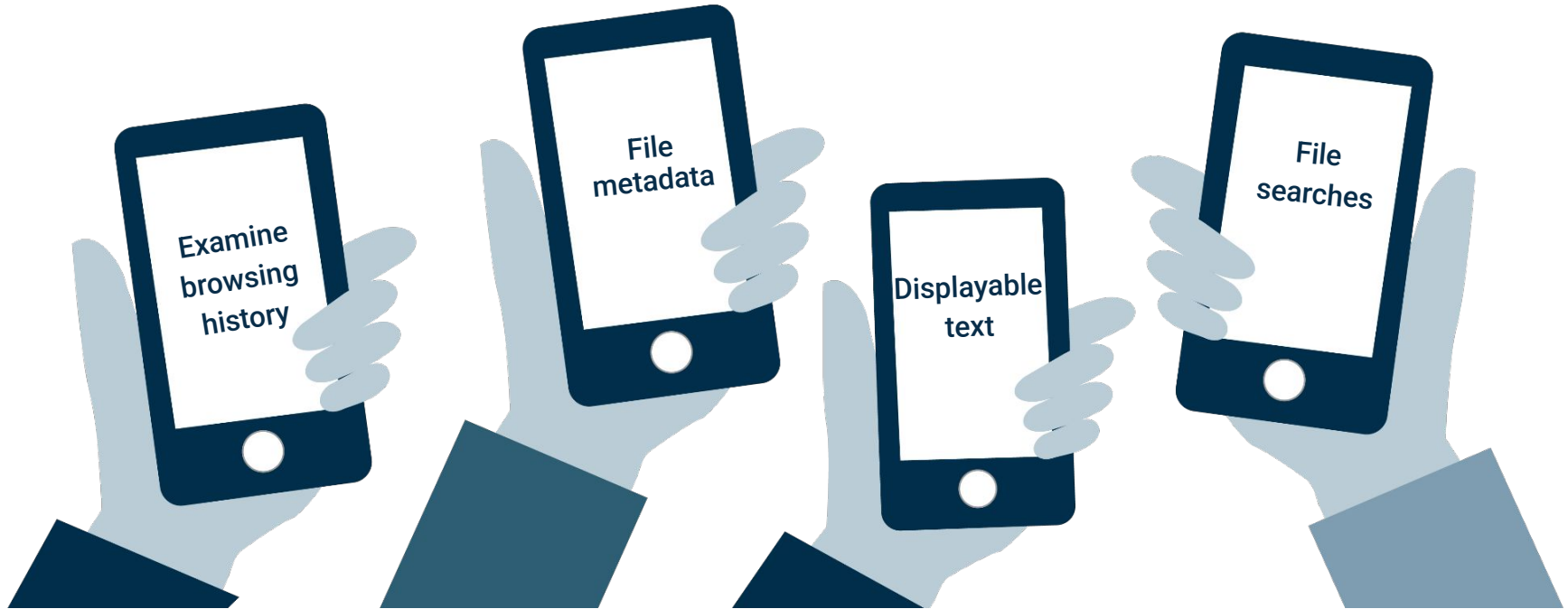
File associated with backup and restore.
Contains an assortment of information including host name and OS version.

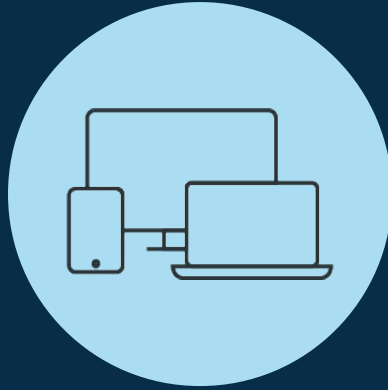




Demonstration Scenario

In this demonstration, we will go through various ways to obtain evidence.





Instructor Demonstration

Evidence analysis with Autopsy



Questions?





Activity

Mobile evidence analysis

In this activity, you will play the role of a digital forensics investigator.

- You are tasked with analyzing evidence and creating a spreadsheet that profiles the details of Peter's iPhone.
- This will serve as your preliminary documentation for the remaining activities.

Suggested Time
50 minutes



Time's Up!
Let's review



Questions?

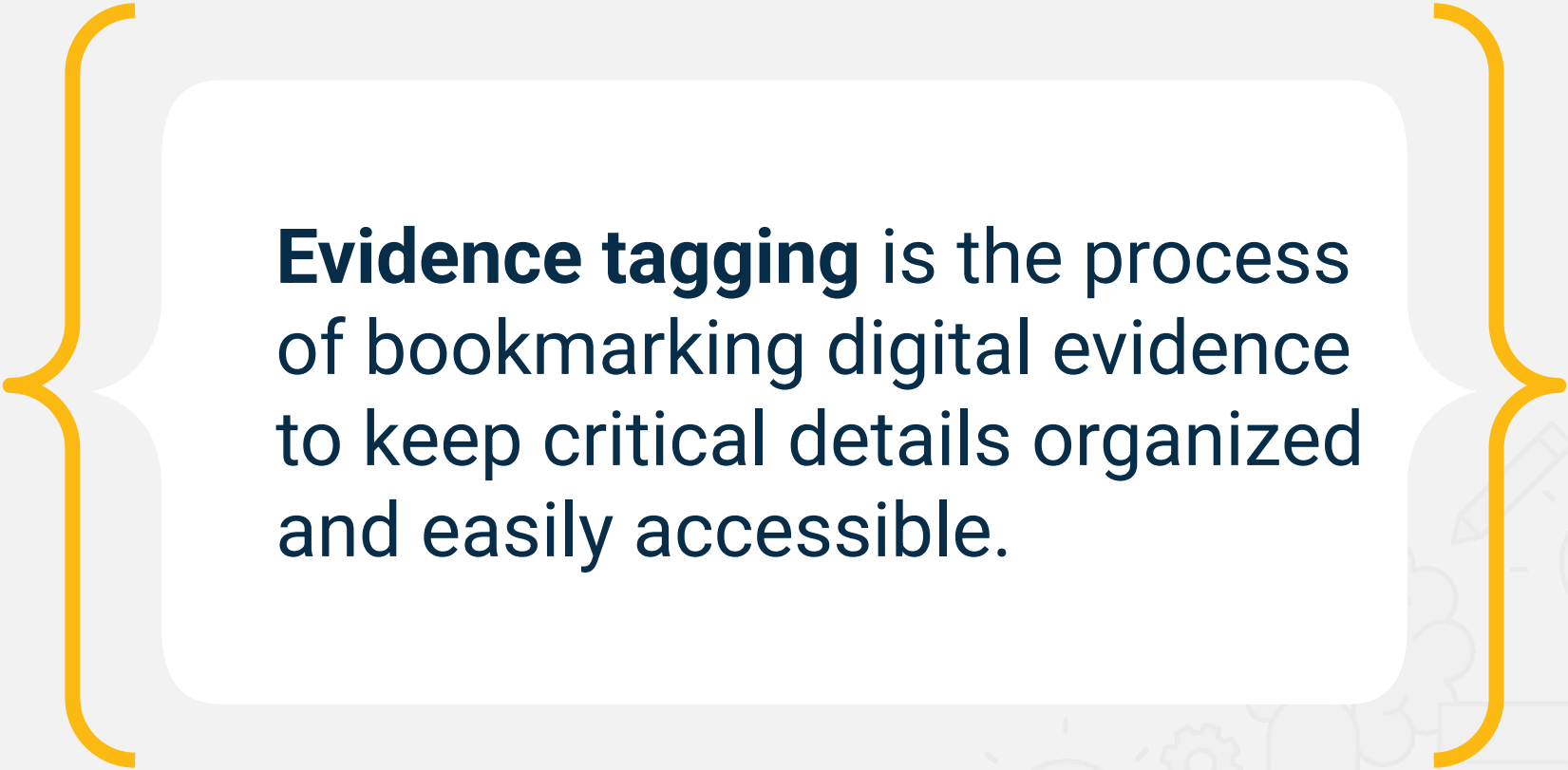




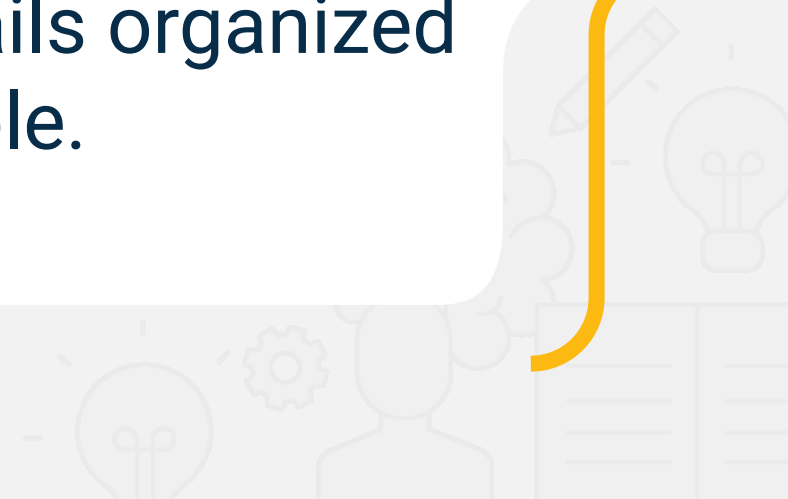
Break
15 mins



Evidence Tagging



Evidence tagging is the process of bookmarking digital evidence to keep critical details organized and easily accessible.



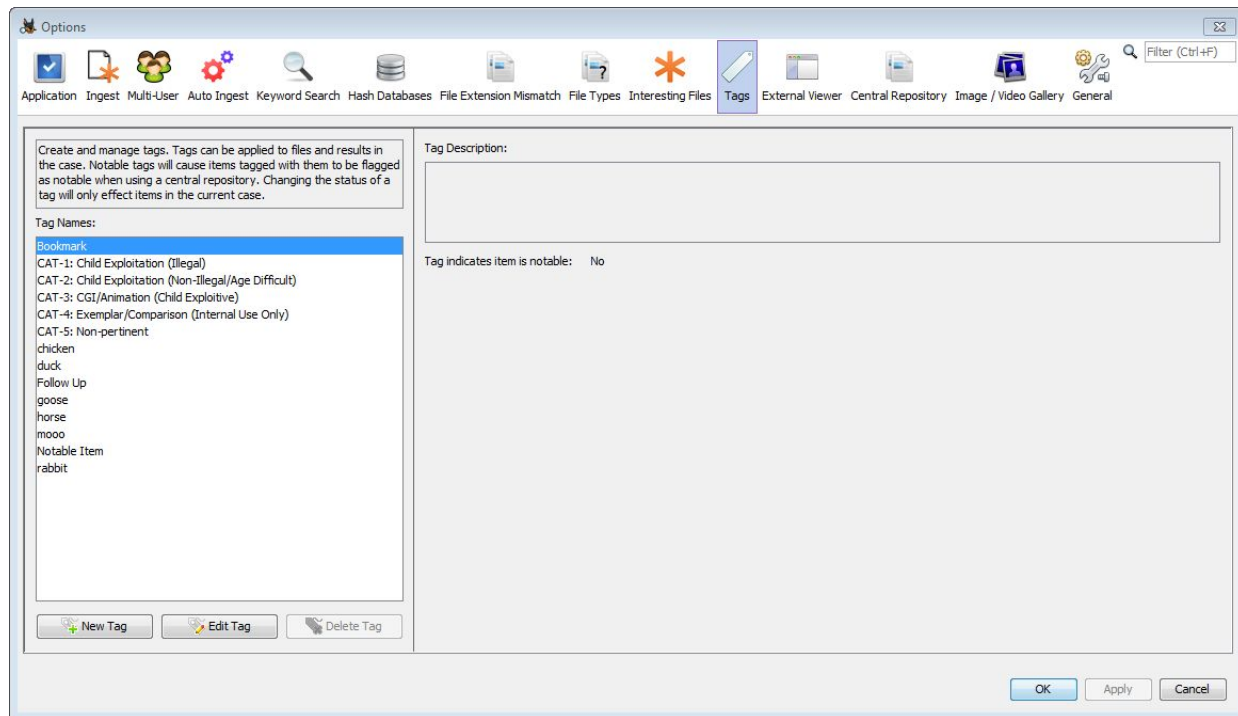


We'll be tagging evidence throughout our investigation to help categorize and organize all digital evidence pertinent to the Pure Gold Credit Union case.

Tagging Evidence

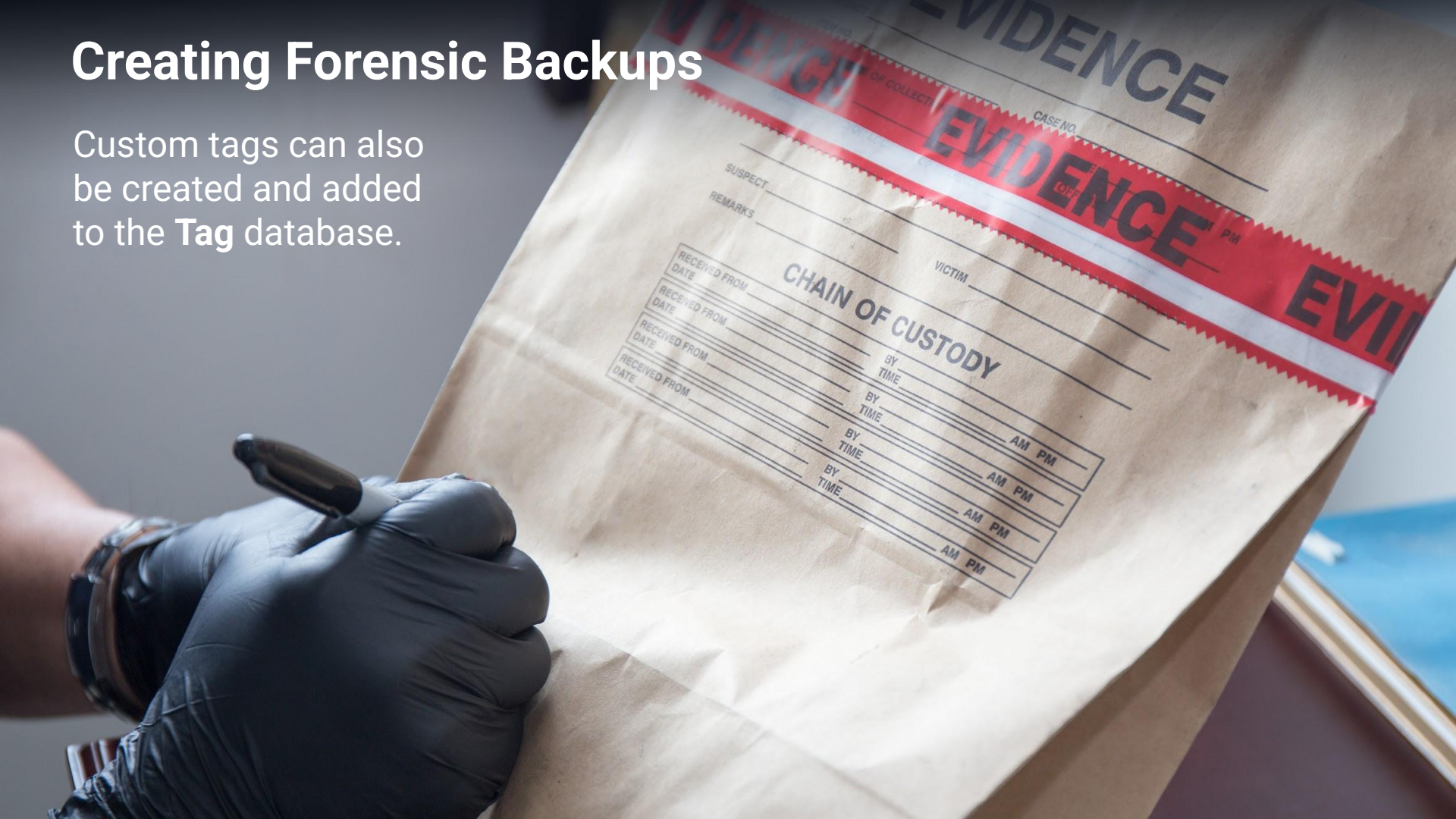
Autopsy includes a built-in feature to tag evidence that contains a set of predefined tags including:

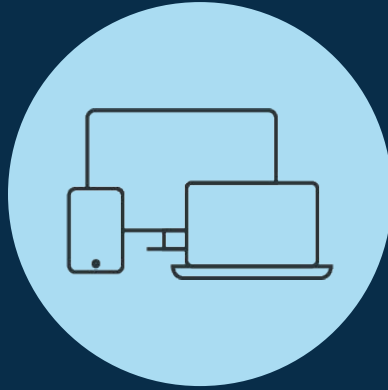
- 1 Follow Up
- 2 Notable Item
- 3 Child Exploitation
- 4 Uncategorized
- 5 Non-Pertinent



Creating Forensic Backups

Custom tags can also be created and added to the **Tag** database.





Instructor Demonstration

Evidence tagging



Activity

Evidence tagging

In this activity, you will tag the major databases and files in the iPhone image file.

- You will create new tags for evidence that is relevant to the case, such as mail, address book, and SMS.
- Be sure to tag as much relevant information as possible. It will be used in later activities as you build your case.

Suggested Time
15 minutes



Time's Up!
Let's review



Questions?





Extracting Data for
Offline Analysis

Sometimes investigators use other applications for further offline analysis.

01

Not all data types can be rendered in Autopsy.

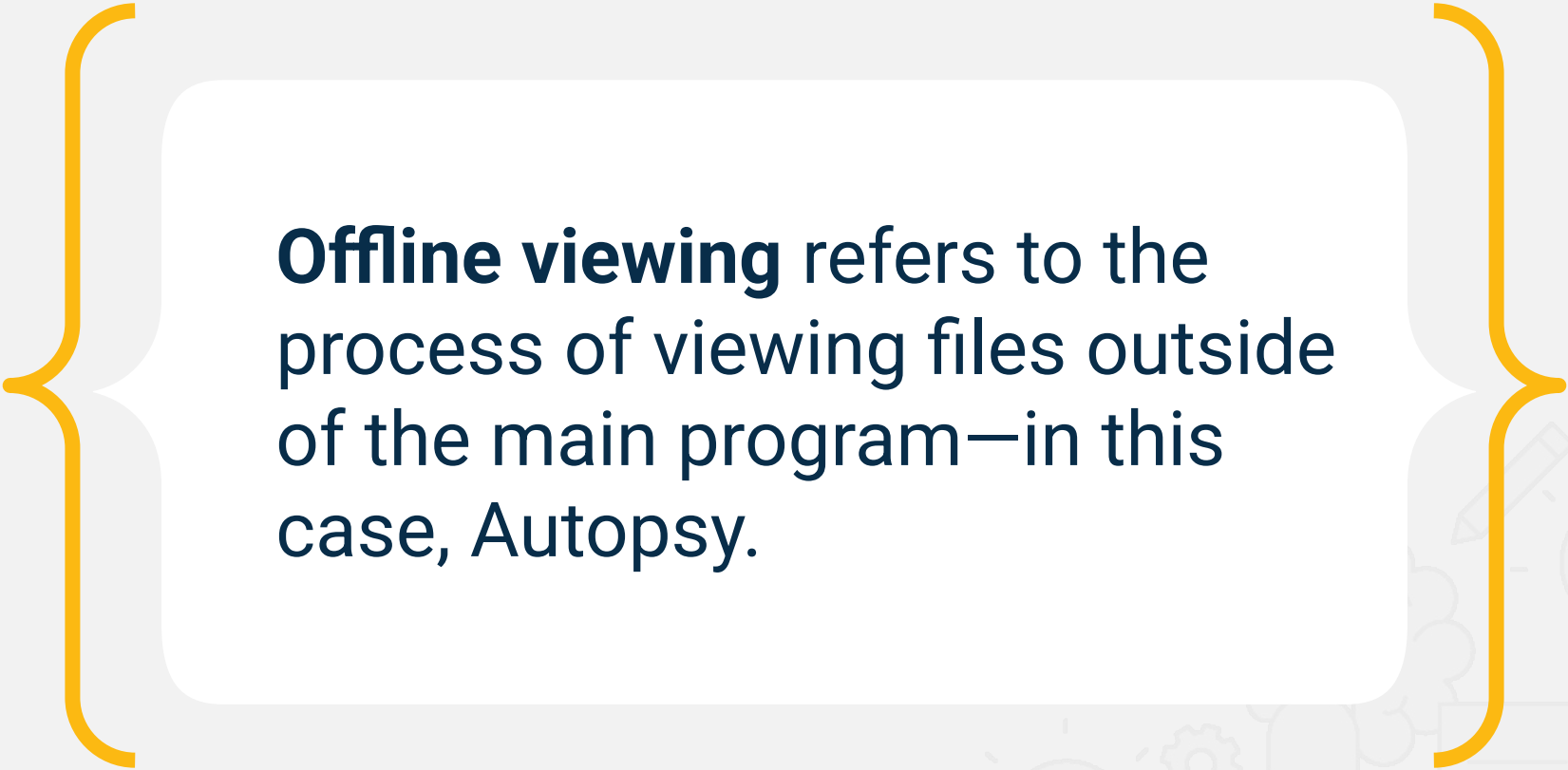
02

Other tools can analyze video, photo, and audio files in more depth.

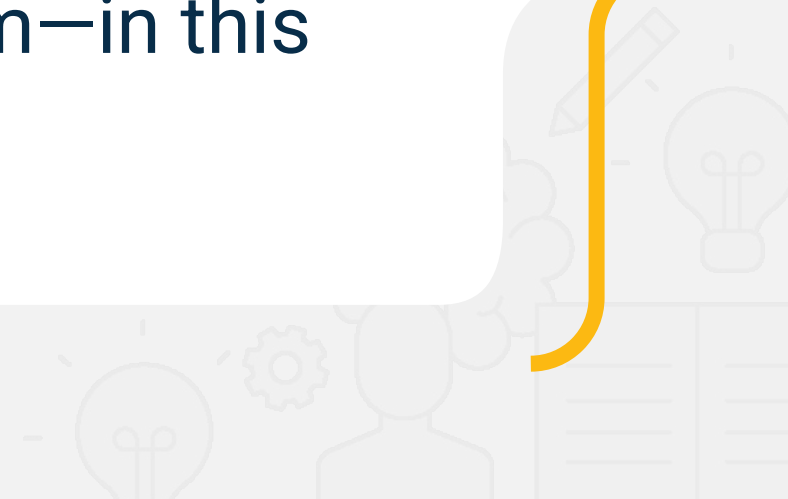
03

Database information can easily be transferred to spreadsheets or Word documents where it can be manipulated into reports.



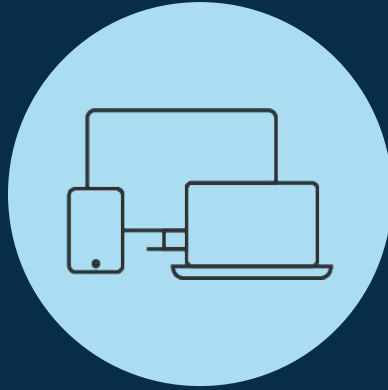


Offline viewing refers to the process of viewing files outside of the main program—in this case, Autopsy.





In the following guided tour,
we'll extract files and directories
for offline viewing.



Instructor Demonstration

Single file export



Activity

Extracting data for offline analysis

In this activity, you will continue in your role as a digital forensics investigator working on the Pure Gold Credit Union case.

- Your task is to create export files for offline examination.

Suggested Time
20 minutes



Time's Up!
Let's review



Questions?





Let's Recap



Recap

Digital forensics

01

As a digital forensics investigator, you will work as part of a team that consists of a large collaborative effort.

02

It's critical, as a digital forensics investigator, to understand how to use tools such as Autopsy and how to export data so that other team members can perform offline analysis of evidence.

03

Tagging evidence helps categorize and label evidence that has already been screened.

04

This eliminates double work and helps other investigative team members continue your work if you become unavailable.



Next Class

We will continue our investigation by analyzing email messages, SMS messages, and web history.



Thank You